Identification and Control of a Propeller-Actuated Damped Pendulum (PADP)

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***Abstract*—This work presents the experimental identification and controller design for a propeller-actuated damped pendulum (PADP). Three data sets were acquired and transfer functions were obtained through parameter approximation and MATLAB's System Identification toolbox. Statistical analysis led to the rejection of the sample affected by disturbances, and the final plant model was computed as the average of the remaining valid models. Four controllers were designed — continuous PID, integral state feedback via LQR, I-PD by pole placement, and discrete PID — and physically validated against settling time and overshoot specifications of 5 s and 3%, respectively. [Continue with results summary...]**

I. Introduction

The propeller-actuated damped pendulum (PADP) is a benchmark system in control engineering that exhibits second-order underdamped dynamics driven by a motor-propeller assembly. Its behavior is governed by the interplay between gravitational restoring force, mechanical damping, and the aerodynamic torque generated by the propeller, making it a representative platform for studying identification and control of oscillatory systems [1].

Accurate plant identification is a prerequisite for effective controller synthesis. Parametric methods extract the natural frequency ω_n and damping ratio ζ directly from measured transient responses, while data-driven tools such as MATLAB's System Identification toolbox fit transfer function models by minimizing a prediction error criterion [2]. Since experimental data are inherently subject to noise and occasional disturbances, statistical screening of the identified models is essential before model averaging is performed.

Once the plant model is established, different control strategies offer distinct trade-offs between performance, robustness, and implementation complexity. The classical PID controller remains the industrial standard due to its simplicity and intuitive tuning [3]. Optimal state-feedback control via the Linear Quadratic Regulator (LQR) provides a systematic way to balance control effort and tracking error, and the addition of an integral state allows rejection of constant disturbances [4]. The I-PD structure — where the integrator acts on the error while the proportional and derivative terms act only on the plant output — eliminates derivative kick and reduces overshoot compared to the standard PID [5]. Finally, discrete-time implementation is mandatory when the controller runs on a digital platform; the choice of sampling period and discretization method directly affects closed-loop stability and performance [6].

This paper reports the complete identification-to-validation cycle for the PADP system. Section II covers experimental data acquisition, parameter approximation, MATLAB-based identification, statistical model selection, and the derivation of the final plant model. Section III presents the design of four controllers. Sections IV and V report and discuss the simulation and experimental results, respectively, evaluated against the design specifications of settling time $t_s \leq 5$ s and maximum overshoot $M_p \leq 3\,\%$.

II. System Identification

A. Experimental Data Acquisition

Three independent experiments were conducted on the PADP to capture the dynamic response of the system under nominal conditions. In each trial, an excitation signal was applied and the angular response of the pendulum was recorded.

B. Parameter Approximation

From the experimental data, the parameters ω_n and ζ of each sample were estimated through transient response analysis, yielding three transfer functions of the form:

$$G_i(s) = \frac{K\,\omega_{n,i}^2}{s^2 + 2\zeta_i\,\omega_{n,i}\,s + \omega_{n,i}^2}, \quad i = 1, 2, 3 \tag{1}$$

C. System Identification via MATLAB Toolbox

Additionally, MATLAB's System Identification toolbox was used to fit a second-order transfer function model to each dataset, yielding three additional models with their respective goodness-of-fit percentages (*best fit*).

D. Statistical Analysis and Sample Rejection

With the six available transfer functions, a statistical analysis of the identified parameters was performed. The results indicated that the TFs corresponding to sample 3 deviated significantly from the distribution of the remaining data, evidencing the presence of a disturbance during that trial. Therefore, sample 3 was discarded from the modeling process.

E. Final Plant Model

The final plant model was obtained by averaging the parameters of the four valid transfer functions:

$$G(s) = \frac{K_{\text{avg}}\,\omega_{n,\text{avg}}^2}{s^2 + 2\,\zeta_{\text{avg}}\,\omega_{n,\text{avg}}\,s + \omega_{n,\text{avg}}^2} \tag{2}$$

III. Controller Design

The design specifications for all controllers are:

- Settling time: $t_s \leq 5$ s
- Maximum overshoot: $M_p \leq 3\,\%$

A. Continuous PID Controller (SISOtool)

A continuous-time PID controller was designed using MATLAB's SISOtool with automatic tuning, based on the identified plant model.

$$C_{\text{PID}}(s) = K_p + \frac{K_i}{s} + K_d\,s \tag{3}$$

B. Integral State Feedback Controller (ISF) via LQR

An Integral State Feedback (ISF) controller was designed using the Linear Quadratic Regulator (LQR) criterion. The method augments the plant with an integral state to ensure zero steady-state error and improve disturbance rejection.

The identified plant in controllable canonical form is:

$$\mathbf{A} = \begin{bmatrix} -0.2481 & -15.4300 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 5.2870 \end{bmatrix}, \tag{4}$$

The system is fully controllable (rank $= 2$). To include integral action, the augmented model is defined as:

$$\dot{x}_I = r - \mathbf{C}\mathbf{x} \tag{5}$$

$$\mathbf{A}_a = \begin{bmatrix} -0.2481 & -15.4300 & 0 \\ 1 & 0 & 0 \\ 0 & -5.2870 & 0 \end{bmatrix}, \quad \mathbf{B}_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{6}$$

The augmented system is also controllable (rank $= 3$). Based on the specifications ($t_s \leq 5$ s, $M_p \leq 3\,\%$), the desired poles were selected as:

$$p_{1,2} = -2.8 \pm 3.1266j, \quad p_3 = -14.0 \tag{7}$$

The LQR weighting matrices were chosen using Bryson's rule:

$$\mathbf{Q} = \text{diag}\left(\frac{1}{9^2}, \frac{1}{7^2}, \frac{1}{0.45^2}\right), \quad R = \frac{1}{0.6^2} \tag{8}$$

The resulting gain vector $\hat{\mathbf{K}} = [k_1 \;\; k_2 \;\; K_i]$ is:

$$\hat{\mathbf{K}} = \begin{bmatrix} 0.7214 & 0.4370 & -1.3333 \end{bmatrix} \tag{9}$$

which yields the closed-loop poles:

$$p_{1,2} = -0.2593 \pm 3.9454j, \quad p_3 = -0.4509 \tag{10}$$

C. I-PD Controller via Pole Placement

The I-PD controller was designed by placing the closed-loop poles at positions that satisfy the performance specifications.

$$C_{\text{I-PD}}(s) = \frac{K_i}{s} - (K_p + K_d\,s) \tag{11}$$

D. Discrete PID Controller (SISOtool)

A discrete-time PID controller was designed using SISOtool with a sampling period of $T = \ldots$ s, employing the $\ldots$ discretization method.

IV. Results and Analysis

This section presents the results obtained in both simulation and physical implementation for each controller, evaluating compliance with the performance specifications ($t_s \leq 5$ s, $M_p \leq 3\,\%$).

A. Continuous PID Controller

B. Integral State Feedback Controller via LQR

The closed-loop step responses of the ISF-LQR controller for $r = 0.6$ rad are shown in Fig. 1 (experiment) and Fig. 2 (simulation). The corresponding performance metrics are summarized in Table I.

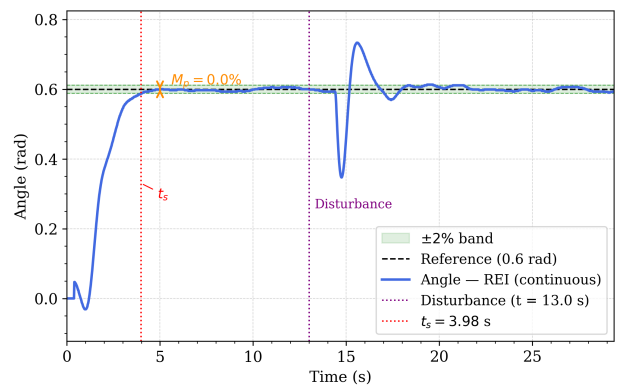

Fig. 1. Experimental step response of the ISF-LQR controller.

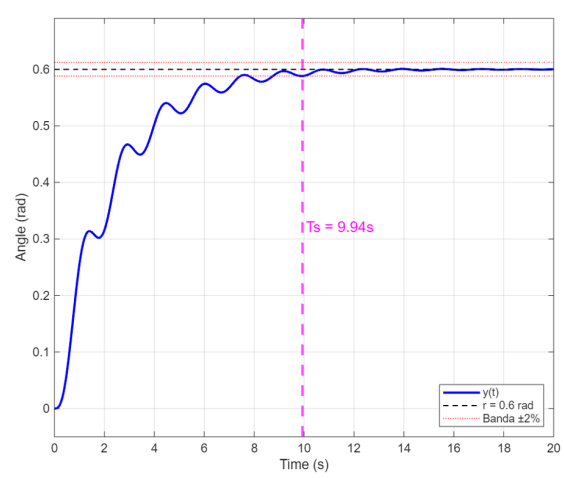

Fig. 2. Simulated step response of the ISF-LQR controller.

TABLE I
ISF-LQR PERFORMANCE METRICS: SIMULATION VS. EXPERIMENT

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	9.94	0.12	0	**No**
Experimental	3.98	0	0	**Yes**

As observed, the simulated response exhibits a settling time of 9.94 s, exceeding the design requirement of 5 s, although the overshoot remains negligible at $0.12\,\%$. In contrast, the experimental implementation achieves a significantly faster settling time of 3.98 s with no observable overshoot, fully complying with the specified performance criteria. Both cases satisfy the steady-state error requirement ($e_{ss} = 0$) due to the inclusion of integral action. Additionally, the overshoot constraint ($M_p \leq 3\,\%$) is met in both simulation and experimental results.

C. I-PD Controller via Pole Placement

D. Discrete PID Controller

E. Comparative Summary

V. Discussion

This section interprets the results presented above, analyzing the causes of the observed differences between theoretical models and the physical behavior of the system, as well as the advantages and disadvantages of each implemented control strategy.

A. ISF-LQR: Simulation–Experiment Discrepancy

Figs. 2 and 1 show a clear contrast between simulated and experimental responses. The simulation exhibits a slowly damped transient with $t_s = 9.94$ s, consistent with the real pole $p_3 = -0.4509$ near the imaginary axis, which dominates the settling behavior despite the faster complex poles. In contrast, the experimental response converges monotonically to the 0.6 rad reference in $t_s = 3.98$ s with zero overshoot, satisfying both specifications.

A small negative deflection at $t \approx 1$ s in the experimental response (about -0.07 rad) is attributed to actuator dead-zone and propeller inertia, which delay the onset of effective control action. This effect is not captured by the linear model.

The faster experimental convergence is consistent with additional nonlinear damping (e.g., friction and aerodynamic drag), which increases the effective damping and suppresses oscillations, leading to a shorter settling time than predicted.

Finally, the experimental response shows effective disturbance rejection; a perturbation at $t \approx 13$ s produces a deviation of approximately 0.25 rad, after which the output returns to the reference, remaining within the margin of less than 2% steady-state error. This confirms the role of comprehensive action in rejecting disturbances despite the model's inaccuracies.

VI. Conclusions

The ISF-LQR controller achieved zero steady-state error and effective disturbance rejection. While the simulation resulted in $t_s = 9.94$ s and $M_p = 0.12\,\%$, failing the settling time requirement, the experimental response reached $t_s = 3.98$ s with $M_p = 0\,\%$, satisfying all specifications.

References

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